| Reg. No | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|

# St. Joseph's College of Arts & Science (Autonomous)

**St. Joseph's College Road, Cuddalore – 607001**

**ASCA202 – STATISTICAL METHODS**

**Time : 3 hrs** **Max Marks :75**

**SECTION - A (10x1=10)**

**Answer ALL the questions**

1. Define Central tendency.
2. What are the types of skewness?
3. What are the properties of correlation co-efficient?
4. What is meant by rank correlation?
5. Define sample space.
6. Define conditional probability.
7. Write any two uses of chi-square statistic.
8. Define standard error.
9. What is meant by large sample test?
10. Define Null and Alternative hypothesis.

**SECTION - B (5x5=25)**

**Answer ALL the questions**

11. a) Calculate Mean from the following data.

| X | 20 | 25 | 30 | 35 | 40 | 45 | 50 |
|---|---|---|---|---|---|---|---|
| F | 3 | 7 | 30 | 42 | 22 | 12 | 4 |

(Or)

b) Compute standard deviation from the data given below.

8, 16, 25, 12, 14, 16, 20, 22

12. a) Explain scatter diagram

(Or)

b) Find rank correlation from the following data.

| MARKS IN STATISTICS | 36 | 48 | 60 | 52 | 90 | 75 | 82 | 40 |
|---|---|---|---|---|---|---|---|---|
| MARKS IN ACCOUNTANCY | 46 | 53 | 75 | 82 | 38 | 43 | 78 | 60 |

13. a) State and prove Addition Theorem on probability.

(Or)

b) The probabilities of X,Y,Z becoming managers are 4/9, 2/9, and 1/3 respectively. The Probabilities that the bonus scheme will be introduced if X,Y,Z becomes managers are 3/10, ½, and 4/5 respectively. What is the probability that the manager appointed was X?

14. a) Explain chi-square test for goodness of fit.

(Or)

b) A random sample of 27 pairs of observation from a normal population gave a correlation co-efficient of 0.6. Is this significant of correlation in the population?

15. a) A sample of 900 members has a mean 3.4cms and standard deviation 2.61cms. Is the sample from a large population of mean 3.25cms and standard deviation 2.61cms?

(Or)

b) Explain testing the significant difference between two proportions.

**SECTION - C (4x10=40)**

**Answer any FOUR Questions**

16. Explain Graphical reqresentation of data.

17. Compute Karl Pearson's Co-efficient of correlation from the following data.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| X | 25 | 32 | 20 | 35 | 30 | 32 | 36 | 38 | 40 |
| Y | 45 | 48 | 32 | 50 | 53 | 56 | 60 | 62 | 55 |

18. State and Prove Baye's Theorem.

19. Two random samples drawn from two normal populations are

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SAMPLE I | 20 | 16 | 26 | 27 | 23 | 22 | 18 | 24 | 25 | 19 | | |
| SAMPLE II | 27 | 33 | 42 | 35 | 32 | 34 | 38 | 28 | 41 | 43 | 30 | 37 |

Test whether the population have same means.

20. a) In one sample of 8 observations the sum of square of deviation from the sample means 84.4 and in the other sample of 10 observations was 102.6. Test whether the difference is significant at 5% level.

b) In an experiment on immunization of cattle from tuberculosis,the following results were obtained.

| | AFFECTED | UN-AFFECTED |
|---|---|---|
| INOCULATED | 12 | 28 |
| NOT INOCULATED | 13 | 7 |

Examine the effect of vaccine in controlling the incidence of the disease.

21. Random samples drawn from two countries gave the following data relating to the height of adult males

| | COUNTRY A | COUNTRY B |
|---|---|---|
| MEAN HEIGHT(INCHES) | 67.42 | 67.25 |
| S.D (INCHES) | 2.58 | 2.50 |
| NUMBER IN SAMPLES | 1000 | 1200 |

Test whether there is any significant difference between the two population means and S.D's.

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